



MINOR PROJECT

RAG-Based University Academic Assistant using Local LLMs

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1. Broad Area of Work

Artificial Intelligence, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Information Retrieval, and Full-Stack Web Application Development.

2. Introduction

Universities maintain academic and administrative information in documents such as syllabi, academic regulations, notices, course materials, timetables, and previous-year question papers. Students and faculty often need to search across multiple documents to locate specific information. Traditional keyword-based searching may require users to know the terminology used in the source documents.

The proposed project, **RAG-Based University Academic Assistant using Local LLMs**, will provide a natural-language interface through which authorized university users can ask questions and receive answers grounded in university documents.

Retrieval-Augmented Generation (RAG) will retrieve relevant document content before providing context to a locally hosted Large Language Model through Ollama. The approach is intended to provide source-based responses while keeping institutional information within a controlled local environment.

3. Literature Survey / Work Done in the Field of Proposed Work

Recent developments in Natural Language Processing and Large Language Models have enabled conversational systems capable of understanding and generating human language. However, general-purpose language models may not contain current or institution-specific information and can produce unsupported responses.

Retrieval-Augmented Generation combines information retrieval with language generation. Relevant passages are retrieved from a knowledge source and supplied to the language model as context. Embedding-based vector search can identify semantically relevant text.

Local LLM runtimes such as Ollama provide a way to run supported open-source language models on local infrastructure, which is useful where privacy and data control are important.

The proposed work applies these concepts to university academic information and combines them with role-based access, document management, chat history, and a web-based interface.

4. Existing Gaps

- Academic information is distributed across multiple documents and locations.
- Keyword-based search may not adequately understand natural-language academic questions.
- General-purpose AI assistants may lack institution-specific documents and may generate unsupported information.
- Sending institutional documents to external AI services can raise privacy and data-control concerns.
- Conventional document repositories generally do not provide a unified conversational interface with evidence from source documents.

5. Objectives of the Proposed Work

1. To develop a university-focused academic assistant capable of answering natural-language queries.
2. To implement a Retrieval-Augmented Generation pipeline for retrieving relevant information from university documents.
3. To integrate a locally hosted Large Language Model using Ollama.
4. To provide source-based responses so users can identify the documents supporting an answer.
5. To implement secure authentication and role-based access for Admin, Faculty, and Student users.
6. To provide document management and conversational history for authorized users.
7. To design a modular architecture that supports future academic assistance features.

6. Proposed Methodologies

The proposed system will be developed as a modular full-stack application consisting of a web frontend, backend API layer, document/RAG processing service, vector storage, database, and local LLM runtime.

1. **Authentication and RBAC:** Implement authentication and role-based permissions for Admin, Faculty, and Student users.
2. **Document Management:** Allow authorized users to upload and manage university documents with metadata such as department, semester, subject, and document type.
3. **Document Processing:** Extract text from uploaded documents, clean it, and divide it into meaningful chunks.
4. **Embeddings and Vector Storage:** Convert document chunks into embeddings and store them in a vector database for semantic retrieval.
5. **RAG Pipeline:** Retrieve relevant chunks for a user's question and provide them as context to the local LLM.
6. **Local LLM Integration:** Use Ollama to run a supported local LLM and generate responses based on retrieved context.
7. **Evidence and Chat:** Return relevant source information and maintain authorized users' chat history.
8. **Testing and Integration:** Test modules, APIs, retrieval quality, authentication, and complete end-to-end flows.

7. Expected Outcome of the Proposed Work

The expected outcome is a working university academic assistant that enables authorized students and faculty to ask questions in natural language and receive answers grounded in uploaded university documents.

The system is expected to provide role-based access, document management, semantic retrieval, local LLM-based response generation, and source-based answers.

Its modular design will provide a foundation for future intelligent academic features.

8. Future Scope of the Work

- Voice-based interaction and text-to-speech support.
- Multilingual academic assistance.
- OCR-based processing of scanned documents and images.
- AI-generated quizzes, flashcards, and personalized study plans.

- Previous-year question paper analysis and topic identification.
- Advanced analytics for administrators and faculty.
- Integration with university timetable, notification, or learning-management systems.
- Deployment on university-managed infrastructure with appropriate scaling and monitoring.

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